BJT DC bias II

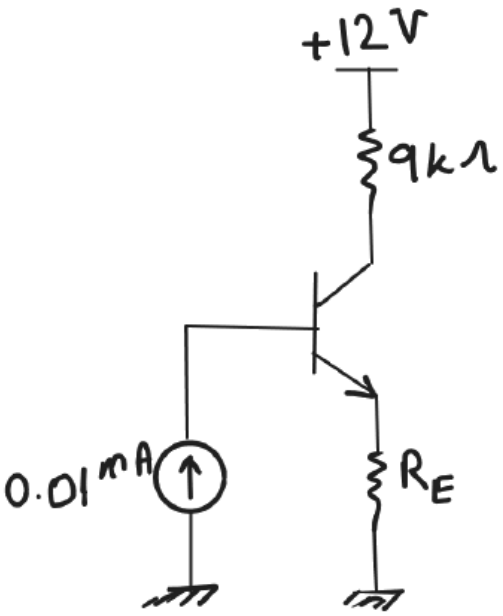
Practice questions · Day 08 · Sat 31 Oct · pack E5

5. Practice

From your course files

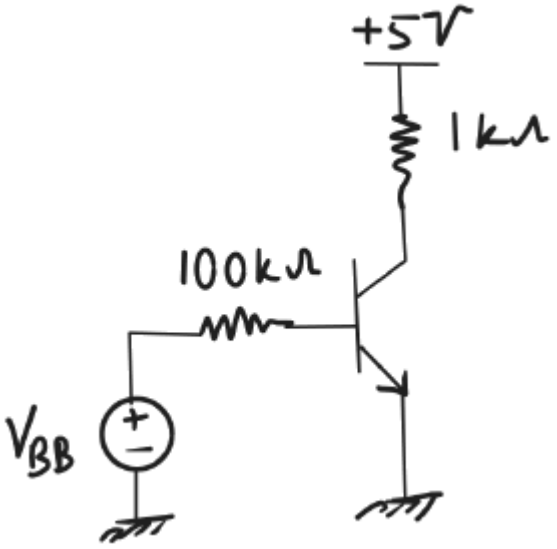
WS5 Q12 (course question). In the circuit below, assume that the BJT has $\beta = 150$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions:

1. For $R_E = 1\ \text{k}\Omega$, find the operating mode of the BJT.
2. Assuming that $100 \leq \beta \leq 200$, for what values of R_E, the transistor will always be in **Saturation** mode?



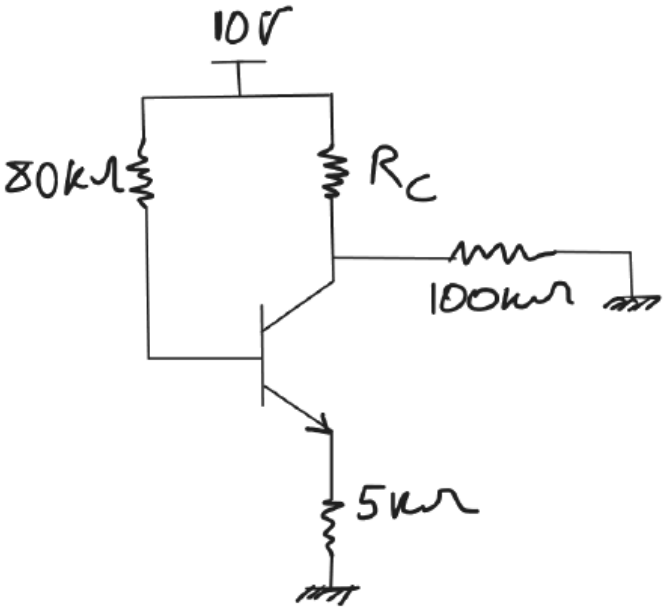
WS5 Q12 figure (from the workshop sheet)

WS5 Q13 (course question). In the circuit below, assume that the BJT has $100 \leq \beta \leq 200$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. What is the minimum base current and base voltage (V_{BB}) in which the transistor will go into **saturation** mode?



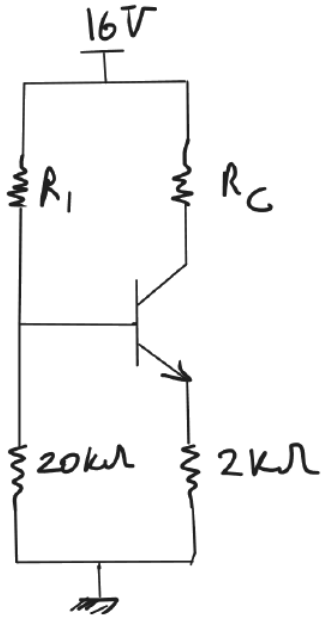
WS5 Q13 figure (from the workshop sheet)

WS5 Q17 (course question). In the circuit below, assume that the BJT has $\beta = 100$, $V_{BE,on} = 0.7$ V, $V_{CE,sat} = 0.2$ V. Find the value of R_C in which $V_{CE} = 1$ V.



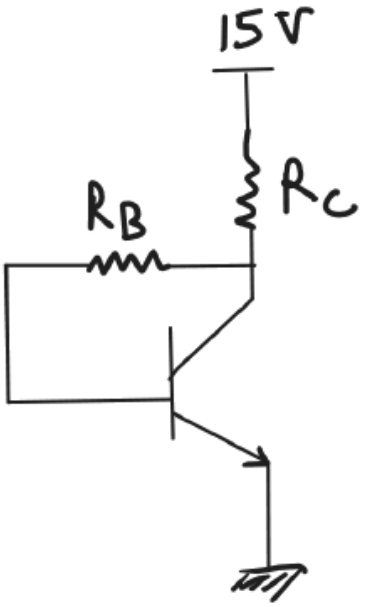
WS5 Q17 figure (from the workshop sheet)

WS5 Q18 (course question). In the circuit below, assume that the BJT has $\alpha = 0.985$, $V_{BE,on} = 0.3$ V, and $V_{CE,sat} = 0.2$ V. Calculate R_1 and R_C to ensure that $I_C = 2$ mA and $V_{CE} = 6$ V.



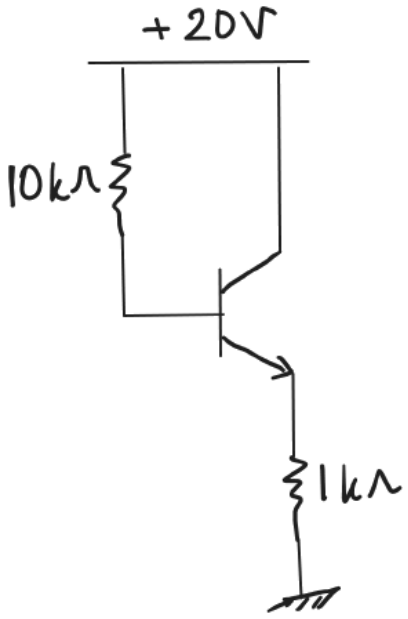
WS5 Q18 figure (from the workshop sheet)

WS5 Q19 (course question). In the circuit below, assume that the BJT has $\beta = 100$ and $V_{BE,on} = 0.3$ V. Calculate R_B to ensure that $I_C = 1$ mA and $V_{CE} = 2$ V.



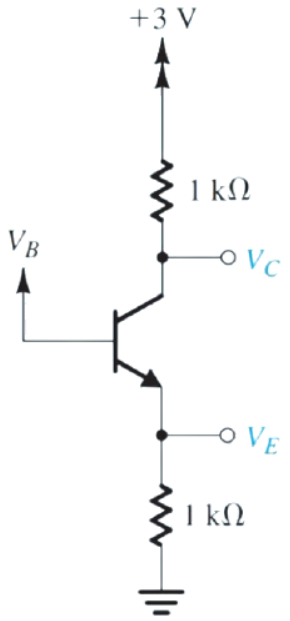
WS5 Q19 figure (from the workshop sheet)

WS5 Q9 (course question, *). In the circuit below, assume that the BJT has $\beta = 100$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the BJT operating mode in this condition.



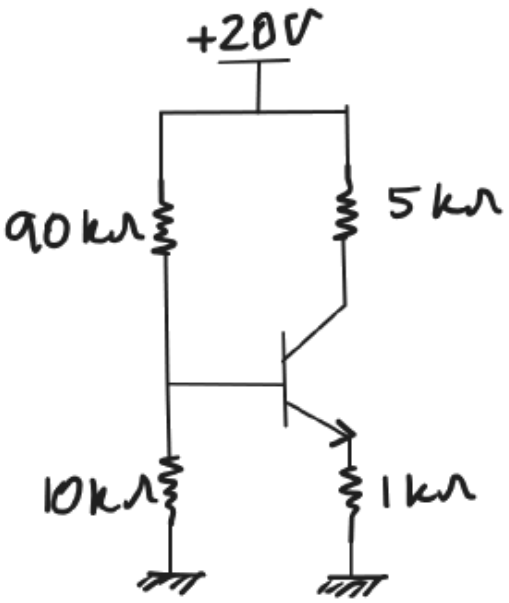
WS5 Q9 figure (from the workshop sheet)

WS5 Q11 (course question, *). In the circuit below, assume that the BJT has a very high β, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. When $V_B = 0$ V, find the BJT operating mode.



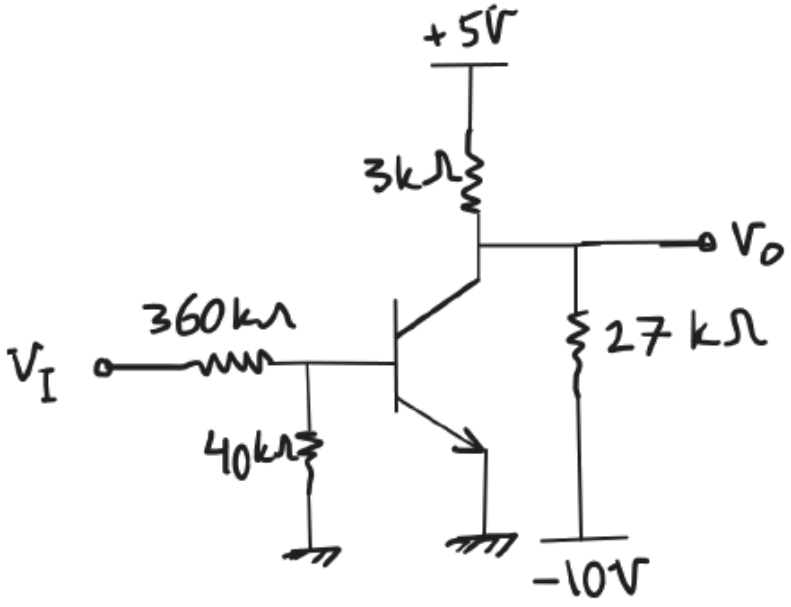
WS5 Q11 figure (from the workshop sheet)

WS5 Q14 (course question, *). In the circuit below, assume that the BJT has $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the base and collector currents and check the transistor operating mode.



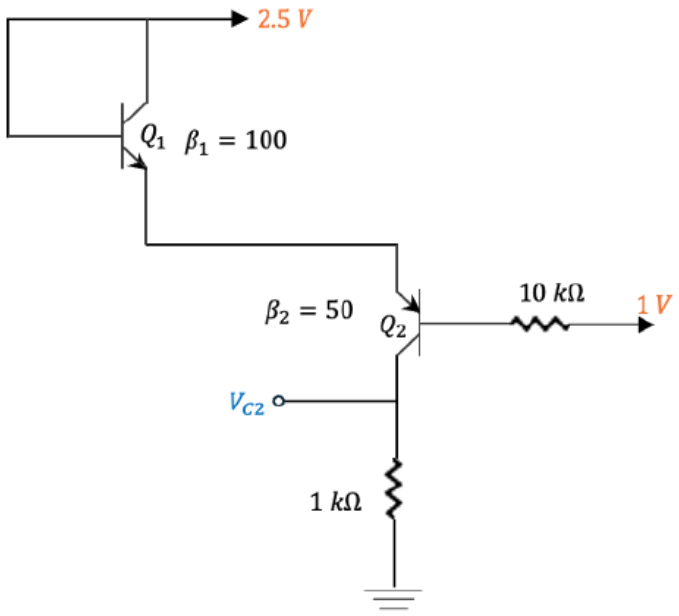
WS5 Q14 figure (from the workshop sheet)

WS5 Q15 (course question, *). In the circuit shown in Figure 1, assume that the BJT has $\beta = 40$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the output voltage, V_o, when $V_I = 15$ V.



WS5 Q15 figure (Figure 1 of the workshop sheet)

PT Q8 (practice test, 6 marks). Consider the circuit below where Q_1 and Q_2 have different current gain, $\beta_1 = 100$ and $\beta_2 = 50$. You can assume $V_{BE,on}$ = 0.7 V and $V_{CE,sat} = 0.2$ V for both BJTs and both BJTs are in **active state**. Find the output voltage, V_{C2} by showing your work.

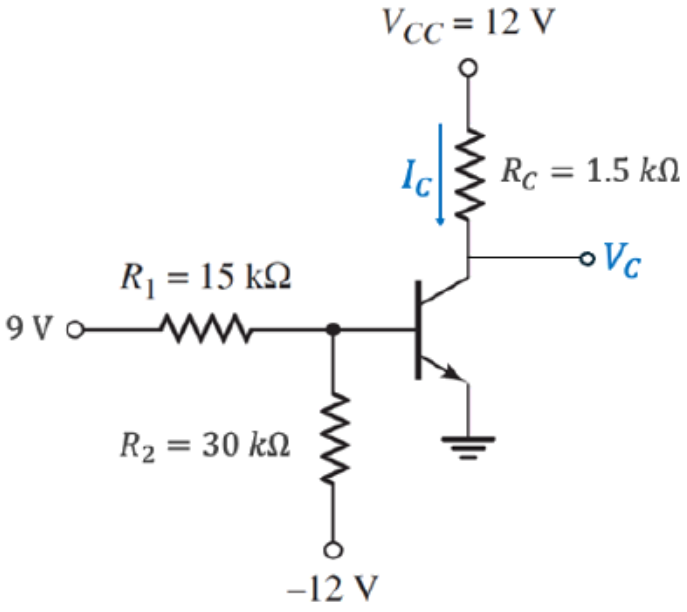


Practice test Q8 figure (from the practice test)

PT Q7 (practice test, 10 marks). For the circuit shown on the right, $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.

a. Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]
b. Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]
c. Determine collector current (I_C) and voltage (V_C). [2 marks]

Then (for this pack): try the approximate divider method on the same circuit and explain what happens.

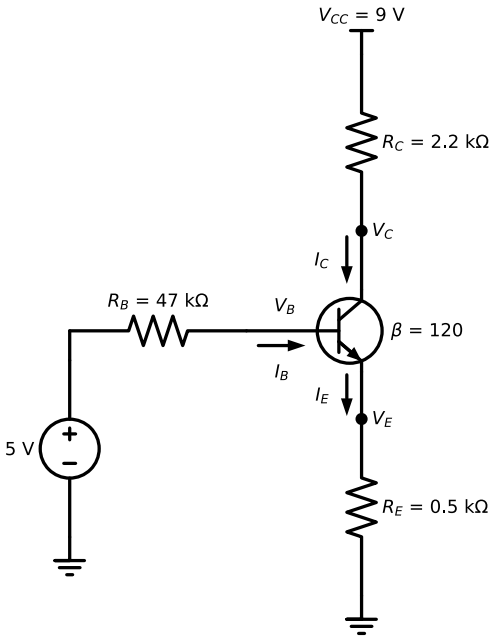


Practice test Q7 figure (from the practice test)

New questions

New Q1 [5 marks]. An npn transistor has its collector connected to V_{CC} = 9 V through R_C = 2.2 kΩ and its emitter to ground through R_E = 0.5 kΩ. Its base is driven from a separate 5 V source through R_B = 47 kΩ. β = 120, $V_{BE,on}$ = 0.7 V, $V_{CE,sat}$ = 0.2 V.

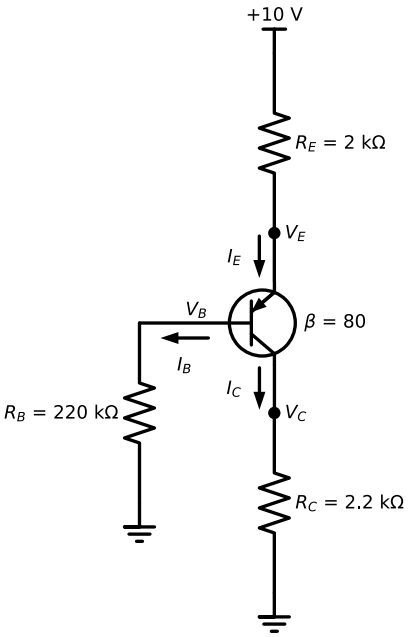
a. Assume active and find I_B, I_C and V_{CE}. [2]
b. Find the true state, I_B and I_C, and the forced β. [3]



Circuit for New Q1: separate 5 V base source with emitter resistor.

New Q2 [6 marks] (pnp). A pnp transistor has its emitter connected to +10 V through R_E = 2 kΩ. Its base is connected to ground through R_B = 220 kΩ. Its collector is connected to ground through R_C = 2.2 kΩ. β = 80, $|V_{EB,on}|$ = 0.7 V, $|V_{EC,sat}|$ = 0.2 V.

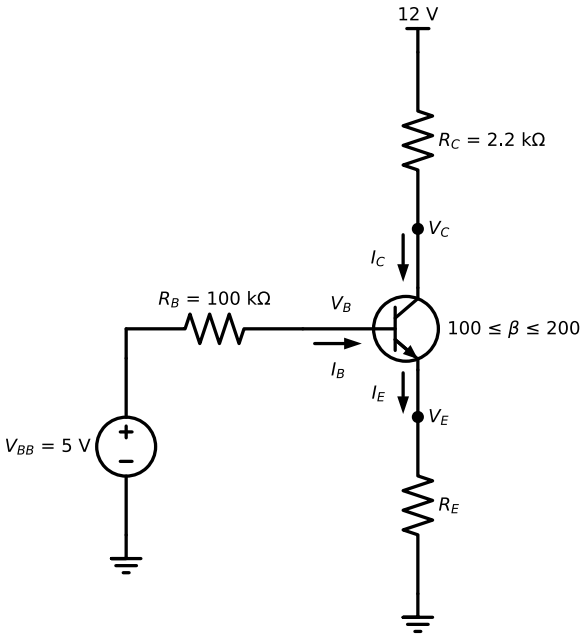
a. Mark the directions of I_B, I_C and I_E, and find I_B and I_C. [3]
b. Find V_E, V_B, V_C and V_{EC}, and confirm the state. [2]
c. What is the largest R_C that keeps the transistor active? [1]



Circuit for New Q2: pnp, emitter 2 kΩ to +10 V, base 220 kΩ to ground, collector 2.2 kΩ to ground.

New Q3 [7 marks] (sensitivity). An npn transistor has its collector connected to 12 V through R_C = 2.2 kΩ. Its base is driven from V_{BB} = 5 V through R_B = 100 kΩ. An emitter resistor R_E connects the emitter to ground. $100 \le \beta \le 200$, $V_{BE,on}$ = 0.7 V, $V_{CE,sat}$ = 0.2 V.

a. With R_E = 0, find I_C and the state at β = 100 and at β = 200. [2]
b. Find the smallest R_E that keeps the transistor active over the whole β range. [3]
c. Choose the next standard value above it (E12) and give the range of I_C and V_{CE}. [2]



Circuit for New Q3: V_BB = 5 V through 100 kΩ, R_E to be chosen.

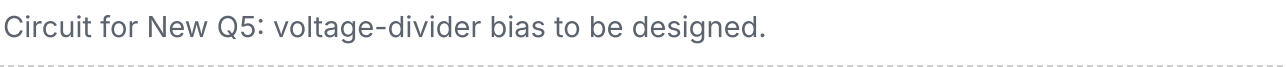
New Q4 [8 marks] (two transistors, npn into pnp). Q_1 is an npn (β1 = 150). Its base is fed by a divider from 12 V: R_1 = 56 kΩ from 12 V to the base and R_2 = 12 kΩ from base to ground. Its emitter goes through R_{E1} = 1 kΩ to ground, and its collector through R_{C1} = 4.7 kΩ to 12 V. Q_1 's collector connects directly to the base of Q_2, a pnp (β2 = 100). Q_2's emitter goes through R_{E2} = 2.2 kΩ to 12 V, and its collector, the output V_{C2}, goes through R_{C2} = 1.8 kΩ to ground. Use $|V_{BE}|$ = 0.7 V and $|V_{CE,sat}|$ = 0.2 V.

a. Test whether the approximate divider method is valid for Q_1, and find I_{C1} by the exact method. [2]
b. Ignoring I_{B2}, find V_{C1}, V_{E2} and I_{E2}. [2]
c. Repeat (b) including I_{B2} (state which way it flows), then find V_{C2} and check both transistors. [3]
d. What is the largest R_{C2} for which Q_2 stays active? [1]

Circuit for New Q4: divider-biased npn Q1 directly driving pnp Q2.

New Q5 [8 marks] (design). Design a voltage-divider bias circuit with V_{CC} = 15 V for I_{CQ} = 2 mA and V_{CEQ} = 6 V. The transistor has $100 \leq \beta \leq 300$.

a. Use $V_E = 0.1V_{CC}$ to find R_E and R_C. [2]
b. Choose R_2 so that the approximate method is valid at $\beta_{\min}$, then find R_1. [2]
c. Choose E24 standard values and use the exact method to find I_{CQ} and V_{CEQ} at β = 100 and β = 300. [4]



Circuit for New Q5: voltage-divider bias to be designed.

New Q6 [6 marks] (collector feedback, β range). An npn transistor has R_C = 3.3 kΩ from 12 V to the collector, R_F = 180 kΩ from collector to base, and R_E = 0.5 kΩ from emitter to ground. V_{BE} = 0.7 V.

a. Find I_C, V_C and V_{CE} for β = 80, 160 and 240. [4]
b. A fixed-bias circuit (R_B from 12 V to the base, the same R_C, no R_E) is designed to give the same I_C at β = 160. Find R_B and the V_{CE} it gives at β = 240. Comment. [2]



Circuits for New Q6: (a) collector feedback, (b) the fixed-bias comparison.

6. Quick recall check

1. An active-mode calculation gives $V_{CE} = -4$ V. What do you assume next, and which two equations replace $I_C = \beta I_B$?
2. You need a transistor to stay active for every β in $100 \leq \beta \leq 200$. Which β do you design for, and why?
3. In a collector-feedback circuit, what current flows through R_C?
4. A pnp has V_E = 3.0 V and is active. What is V_B, and which way do I_B and I_C flow?
5. What two rules of thumb set R_E and R_2 in voltage-divider bias design?